

A Closed-Loop Self-Enriching Hybrid RAG and Tokenized Support Architecture for Massive-Scale Academic Administration

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ABSTRACT

Higher education administrative systems in developing nations encounter severe operational bottlenecks when serving millions of distributed students across thousands of affiliated institutions. At the National University of Bangladesh (administering 3.2 million students across 2,250+ affiliated colleges), student information retrieval and problem resolution have historically suffered from fragmented portal architectures, multilingual query drift (Bengali, English, and phonetic Banglish), high human overhead, and severe hallucination risks from generic Large Language Models (LLMs). This paper presents the design, implementation, and systems-level evaluation of the **National University AI Assistant (NU AI)**, an end-to-end intelligent administrative platform built on Google Gemini, ChromaDB, SQLite, and FastAPI. The system introduces three primary architectural contributions: (1) a **Dual-Tier Hybrid Retrieval Engine** combining an in-memory SQL fast-path cache with 768-dimensional dense vector embeddings and ISO 8601 chronological date filtering across 21,555+ official circulars; (2) an autonomous, closed-loop **Self-Enriching Knowledge Pipeline (gap_queue)** that captures unanswered or low-confidence queries ($\tau < 0.60$), synthesizes candidate answers via multi-model cascades, and incorporates human-in-the-loop verification without requiring LLM parameter retraining; and (3) a **Role-Isolated Support Token Subsystem** with an automated ticket-to-knowledge pipeline that anonymizes resolved student cases and ingests them into vector memory. Empirical benchmarking demonstrates a latency reduction from 10,580 ms to 2.5 ms (4,232x speedup) for routed intent queries, a Time-to-First-Token (TTFT) under 1,200 ms for generative RAG streams, and 100% elimination of authoritative portal URL hallucinations.

Index Terms— Retrieval-Augmented Generation (RAG), Autonomous AI Agents, Academic Service Automation, Vector-Symbolic Hybrid Search, Closed-Loop Continual Learning, Tokenized Issue Resolution, Latency Optimization.

I. INTRODUCTION & PROBLEM STATEMENT

Public higher education governance in developing economies operates under extreme administrative scale and infrastructural asymmetry. The National University of Bangladesh (NU) is the largest affiliating university in South Asia, managing academic curricula, examinations, admissions, and certifications for over 3.2 million students across 2,250+ colleges spanning all 64 administrative districts. The university's digital presence is distributed across four distinct, unintegrated web portals: Student ERP (103.113.200.68/nu-app), Central Web (nu.ac.bd), Online Admission (app11.nu.edu.bd), and Examination Management System (ems.nu.ac.bd).

While commercial Large Language Models possess strong general reasoning, their direct deployment in public institutional environments fails due to temporal blindness, date hallucinations, and multilingual transliteration drift. This work presents NU AI, an open-architecture, production-deployable academic assistant designed for zero-hallucination institutional governance.

II. RELATED WORK

Standard Retrieval-Augmented Generation (RAG) frameworks retrieve relevant document chunks using cosine similarity over dense vector embeddings. However, dense retrieval alone struggles with exact circular identifiers and lacks chronological prioritization. While fine-tuning offers parameter adaptation, it requires high compute and risks catastrophic forgetting. NU AI introduces non-parametric closed-loop enrichment and dual-tier hybrid caching to overcome these fundamental limitations.

III. SYSTEM ARCHITECTURE & IMPLEMENTATION

The NU AI platform consists of five primary subsystems: (1) Politeness-Controlled Distributed Web Crawler (*crawler/*), (2) Dual-Tier Hybrid Retrieval Engine (*backend/rag_engine.py*), (3) Asynchronous Orchestration & LLM Streaming Gateway (*backend/app.py*), (4) Role-Isolated Support Token Engine (*token_service/*), and (5) Closed-Loop Self-Enrichment Worker (*enrichment/*).

A. Politeness-Controlled Distributed Web Crawler

As implemented in *crawler/base_scraper.py* and *crawler/scheduler.py*, the crawler monitors official portals at 4-hour intervals using MD5 content hashing ($H(c) = MD5(c_{clean})$) for change detection, a 1.0s politeness delay, and automated PDF table extraction via *pdfplumber*.

B. Dual-Tier Hybrid Retrieval Engine & Scoring Formulation

Candidate circulars are ranked using a convex combination of dense vector similarity, BM25 lexical relevance, and exponential chronological recency decay:

$$S(d, q) = w_1 \cdot \text{Sim}_{\cos}(e_q, e_d) + w_2 \cdot \text{Score}_{\text{BM25}}(q, d) + w_3 \cdot \Phi(\Delta t) \\ \Phi(\Delta t) = \exp\left(-\lambda \cdot \left(t_{\text{curr}} - t_{\text{pub}}(d)\right)\right)$$

Where $e_q, e_d \in \mathbb{R}^{768}$, $\lambda = 0.00274 \text{ day}^{-1}$ (1-year half-life), and $w_1=0.45, w_2=0.35, w_3=0.20$.

IV. THE CLOSED-LOOP SELF-ENRICHING KNOWLEDGE PIPELINE

As implemented in *enrichment/worker.py*, queries with retrieval confidence $C(q) < 0.60$ are pushed to the *gap_queue* SQLite table. The asynchronous daemon executes multi-model generation cascades across Gemini flash variants, formats candidate answers with official portal citations, and stores drafts in *faq_entries*. High-confidence answers (≥ 0.95) or admin-approved entries are vectorized dynamically into ChromaDB with zero server downtime.

V. ROLE-ISOLATED SUPPORT TOKEN SUBSYSTEM

The ticketing engine (*token_service/*) implements a 6-stage finite state machine (PENDING $\rightarrow$ ASSIGNED $\rightarrow$ PROCESSING $\rightarrow$ SOLVED $\rightarrow$ CLOSED / REJECTED). When a departmental solver resolves an issue, *_index_solved_token()* scrubs Personally Identifiable Information (registration numbers, names, phones) and indexes the resolution into ChromaDB as an anonymized verified case.

VI. EMPIRICAL BENCHMARKS & SYSTEMS EVALUATION

Execution Step / Trace	Baseline Sequential	Optimized Parallel & Stream	Speedup Factor
Greeting / Shortcut Query	10,580 ms	2.5 ms	4,232x Faster
Token Status Query	8,200 ms	15.0 ms	546x Faster
Recent Notices Query	4,300 ms	18.0 ms	238x Faster
Complex RAG Retrieval Stage	7,089 ms (2 remote calls)	450 ms (Single call + SQL)	15.7x Faster
Complex RAG Time-to-First-Token	10,580 – 25,070 ms	850 – 1,200 ms	12x – 25x Faster
Complex RAG Total Stream Time	18,500 – 50,000 ms	1,600 – 2,400 ms	10x – 20x Faster

VII. CONCLUSION & FUTURE DIRECTIONS

NU AI demonstrates that combining dual-tier hybrid caching, chronological ISO date decay, non-parametric closed-loop self-enrichment, and role-isolated support ticket triage produces a high-throughput, zero-hallucination institutional assistant capable of serving 3.2 million university students with sub-second response times.

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